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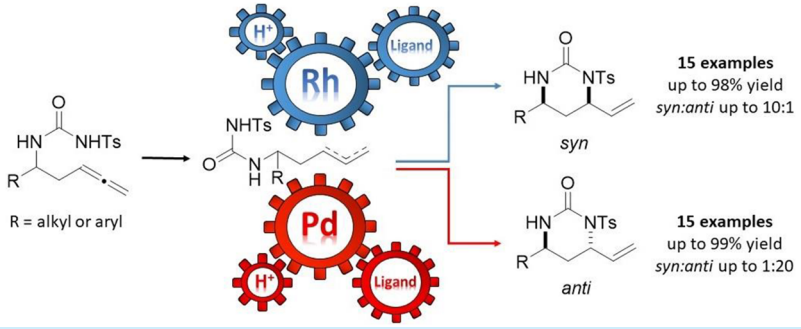
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Stereodivergent Palladium- and Rhodium-Catalyzed Intramolecular Addition of Tosylureas to Allenes: Diastereoselective Synthesis of Tetrahydropyrimidinones

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Cite This: *Org. Lett.* 2021, 23, 9168–9172 Read Online

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ABSTRACT: The intramolecular addition of tosylureas to allenes is highly *syn*-/*anti*-diastereoselective when employing a palladium or rhodium-based catalytic system and affords 1,3-cyclic ureas. Under palladium catalysis a range of thermodynamic *anti*-tetrahydropyrimidinones are accessible, while rhodium catalysis allows synthesis of the kinetic *syn*-tetrahydropyrimidinones. For a representative scope of substrates both cyclic ureas were obtained in excellent yields and diastereoselectivities. The obtained tetrahydropyrimidinones were shown to be easily deprotected and modified to demonstrate the synthetic value.

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Tetrahydropyrimidinones and the related 1,3-diamines are a common motif in organic chemistry and frequently observed in natural products,[1] drugs,[2] and ligands.[3] In contrast to traditional synthetic approaches,[4] more efficient and direct catalytic methods for the synthesis of tetrahydropyrimidinones, which can be conveniently transformed into a 1,3-diamine, were recently discovered. Menche reported a Pd-catalyzed intramolecular allylic substitution of allylcarbonates, resulting in 1,3-disubstituted tetrahydropyrimidinones.[5] Variation of the solvent allowed stereodivergent synthesis of *syn*- and *anti*-tetrahydropyrimidinones in excellent yields and diastereoselectivities. A powerful isomerization/amination method was published by White, who applied Pd-catalyzed C−H-oxidation in allylic position to access 1,2-imidazolidinones.[6] This concept was extended to carbamides by Hara.[7] The use of White's $Pd(TFA)_2$/bis-sulfoxide catalyst and stoichiometric oxidant led to 1,2-imidazolidinones and 1,3-tetrahydropyrimidinones.

In the past decade, methods for the catalytic addition of pronucleophiles to alkynes and allenes have been developed as an atom economic[8] alternative to the allylic substitution.[9] Earlier studies in this field showed that the addition of *N*-pronucleophiles to allenes by late transition metal catalysis leads to valuable branched allylic products in excellent yields and selectivities.[10,11] Very recently we reported the intramolecular enantioselective addition of tosylamides to terminal and internal allenes.[12] Yet the necessity of chiral ligands and matched/mismatched situations still prove to be challenging. This problem was addressed in previous work by our group. By taking advantage of the relative configuration in the product, different catalytic systems were developed for the intramolecular diastereoselective addition of amides[13] and carbamates[14] to allenes. These methods encouraged us to investigate the herein reported intramolecular addition of tosylureas to allenes.

Cyclization of **1a** in the presence of a palladium catalyst and phenylisobutyric acid gave an 85% combined yield of a 1:1 mixture of both diastereomers **2a** and **3a** (Table 1, entry 1). A change of additive toward a decreased amount of benzoic acid improved the yield to 99% and diastereoselectivity to 20:1 in favor of **2a** (entry 2).[15] With a suitable method for the preparation of the *anti*-diastereomer in hand, we conducted a broad screening of different conditions, additives, and catalytic systems in favor of *syn*-selectivity.[15] A major amount of **3a** was observed exclusively under rhodium catalysis. Using PPTS/

Received: October 14, 2021
Published: November 22, 2021



ACS Publications
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https://doi.org/10.1021/acs.orglett.1c03482
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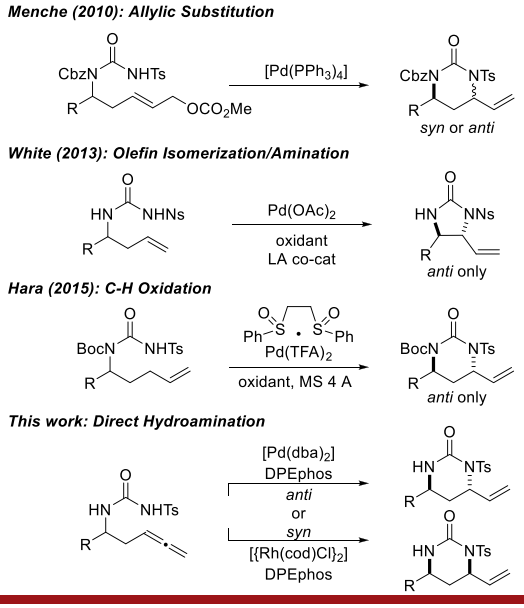
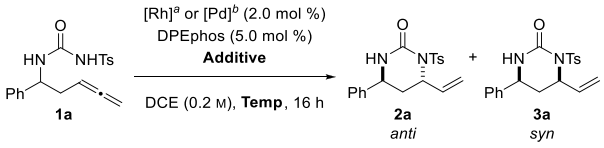


Table 1. Optimization of the Reaction Conditions for the *anti*- and *syn*-Selective Cyclization



No.	Temp (°C)	Catalyst	Additive	yield[c] (%)	d.r.[d] *anti*/*syn*
1	80	[Pd][a]	$PhMe_2CCO_2H$ (100 mol %)	85	1:1
2	80	[Pd][a]	Benzoic Acid (50 mol %)	99	20:1
3	80	[Rh][b]	PPTS (30 mol %) LiCl (30 mol %)	22	1:2
4	80	[Rh][b]	Benzoic Acid (50 mol %)	83	1:3
5	80	[Rh][b]	Pentafluorobenzoic Acid (20 mol %)	99	1:5
6	50	[Rh][b]	Pentafluorobenzoic Acid (20 mol %)	59	1:8
7[e]	50	[Rh][b]	Pentafluorobenzoic Acid (20 mol %)	95	1:10

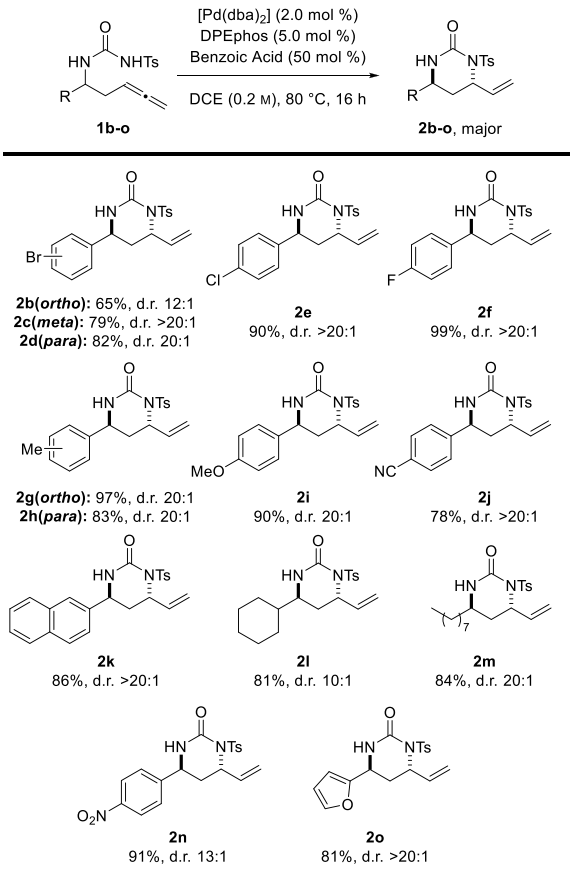
[a]$[Pd(dba)_2]$. [b]$[\{Rh(cod)Cl\}_2]$. [c]Isolated yield. [d]Determined by ^1H NMR of the crude product. [e]Increased catalyst loading to $[\{Rh(cod)Cl\}_2]$ (4 mol %) and DPEphos (10 mol %).

LiCl as additive, which worked fine in the cyclization of carbamates, gave a poor yield and selectivity (entry 3). For benzoic acid an increase of *syn*-selectivity was observed, as well as an increase in yield to 83%. In an extensive screening pentafluorobenzoic acid was found to give an excellent yield and a good diastereoselectivity of 5:1 (entry 5).

A decrease in temperature to 50 °C led to an increase in diastereoselectivity, yet incomplete conversion (entry 6). Increasing the catalyst load at 50 °C pushed the yield to 95%, while maintaining a high diastereoselectivity of 10:1 in favor of **3a** (entry 7).

With suitable methods for both diastereomers in hand, a scope of substrates were tested in *syn*- and *anti*-diastereoselective cyclization. Under palladium-catalysis conditions we achieved an average yield of 86% and average d.r. of 18:1 (Scheme 1).[16] We were able to show that different halogenated aryl groups were tolerated. The *ortho*-brominated substrate **1b** gave a slightly lower yield of **2b**, probably due to side-reactions.

Scheme 1. Scope for the Intramolecular [Pd]-Catalyzed *anti*-Selective Cyclization of Tosylureas[a,b,c]



[a]Isolated yield. [b]d.r. (*anti*:*syn*) determined by ^1H NMR of the crude product. [c]The obtained major isomer is shown.

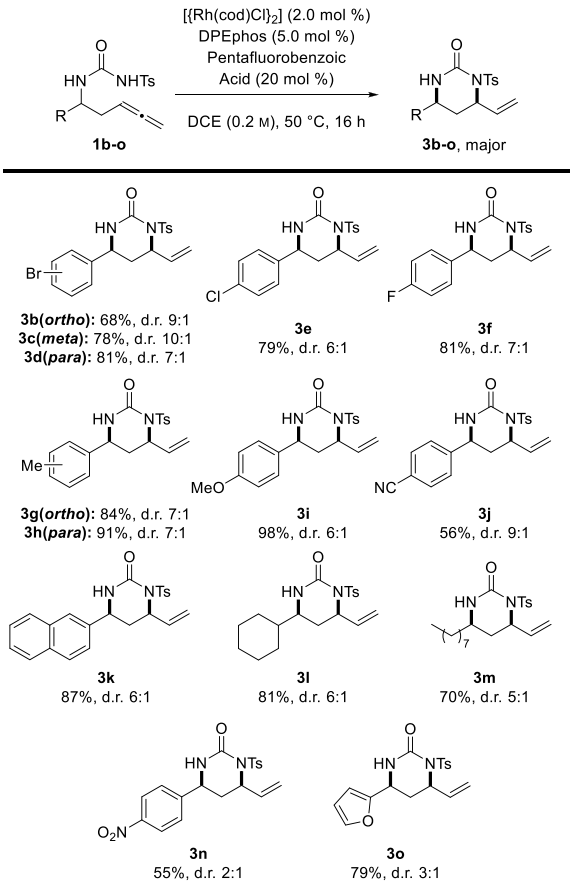
2e and **2f** were synthesized in excellent yield and selectivity. Electron-rich functions in **1g**, **1h**, and **1i** were well tolerated just as the electron-withdrawing functionality in **1j**. Furthermore, steric demands in the naphthyl-derivative **1k** were found to be unproblematic, giving **2k** in 86% yield. Cyclization of aliphatic **1l** and **1m** worked in good yields with a slight drop in d.r. to 10:1 for **2l**. The nitro-containing **2n** was obtained with a d.r. slightly below the average. Performing catalysis on the heteroaromatic furyl-substrate **1o** gave a pleasing 81% yield of **2o**.

Rhodium catalysis allowed access to the valuable *syn*-tetrahydropyrimidinones **3a**–**o** in an average yield of 76% and average d.r. of 7:1 (Scheme 2).[16] For the halogenated aryl-derivatives **1b**–**f** good yields of 68–81% and a d.r. of 6:1 to 10:1 were achieved. In contrast, the electron-rich **2g**–**i** were obtained in slightly higher yield and slightly lower diastereoselectivity. The electron-withdrawing nitro- and cyano-groups in **1j** and **1n** led to only a moderate yield of **3j** and **3n**. Especially **3n** was obtained with a major contamination of the *anti*-isomer **2n**. The aliphatic **3l** and **3m** were synthesized in good yields and moderate diastereoselectivity. **1o** gave a 79% yield of **3o**, yet with a moderate diastereoselectivity of 3:1.

A set of exemplary follow-up reactions have been performed on **3a** (Scheme 3). Reductive cleavage of the tosyl group gave **4** in quantitative yield, and consequent treatment with aqueous HCl allows access to the valuable 1,3-diamine **5**-hydrochloride.[5] Hydrogenation of the double bond affords aliphatic

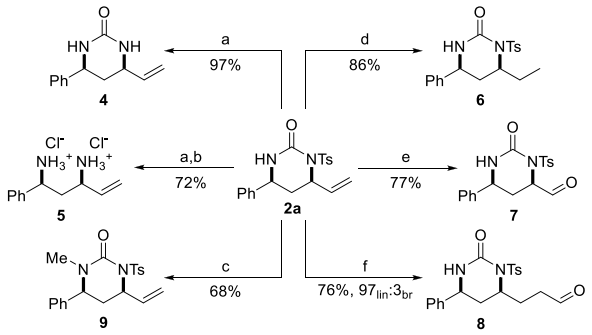
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Org. Lett. 2021, 23, 9168–9172

Scheme 2. Scope for the Intramolecular [Rh]-Catalyzed *syn*-Selective Cyclization of Tosylureas[a,b,c]



[a]Isolated yield. [b]d.r. (*syn*/*anti*) determined of the ^{1}H NMR from the crude products. [c]The obtained major isomer is shown.

Scheme 3. Follow-up Chemistry[a]



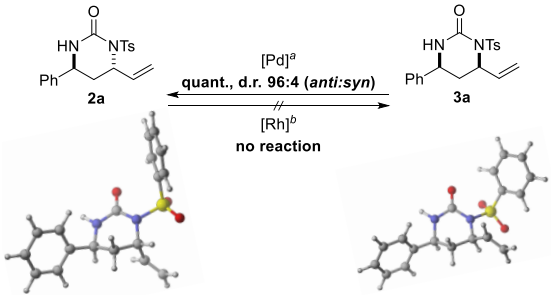
[a]Reaction conditions: (a) Mg-powder (20.0 equiv), MeOH (0.05 M), rt, 16 h, 97%. b) aq. 2 M HCl (10 mL/mmol), 100 °C, 3 h, 72% over two steps. (c) K_2CO_3 (5.0 equiv), Me_2SO_4 (1.5 equiv), MeCN (0.05 M), 70 °C, 16 h. (d) H_2 (10 bar), Pd (5% on charcoal, 10 w %), rt, 18 h, 86%. (e) O_3, MeOH/DCM (5:1, 0.025 M), −78 °C, 10 min then TPP (5 equiv), rt, 1 h, 77%. (f) H_2/CO (1:1, 20 bar), [$Rh(CO)_2$acac] (0.5 mol %), 6-DPPon (5 mol %), toluene (0.025 M), 80 °C, 18 h, 76%.

6 in 86% yield. Ozonolysis and reductive workup with triphenylphosphine allow selective conversion of the olefin group into aldehyde **7**. Employing a self-assembly Rh/6-DPPon-based catalyst system[17] allowed hydroformylation of the olefin function in excellent linear to branched selectivity and 76% isolated yield of **8**. The free urea N−H bond can be directly addressed in functionalization experiments as was demonstrated in the synthesis of alkylated **9**.

When isolated **2a** was subjected to Rh-catalysis conditions, no change in diastereomeric ratio was observed. However, after subjecting **3a** to palladium-catalysis conditions, almost quantitative corrosion of the allylic stereocenter in favor of the *anti*-isomer **2a** was observed after 16 h.

This observation is in accordance with conducted DFT-calculations (BP86/dev2svp), predicting **2a** to be 2.4 kcal more stable than **3a**, therefore **2a** being the thermodynamic and **3a** being the kinetic product of the described cyclization (Scheme 4).

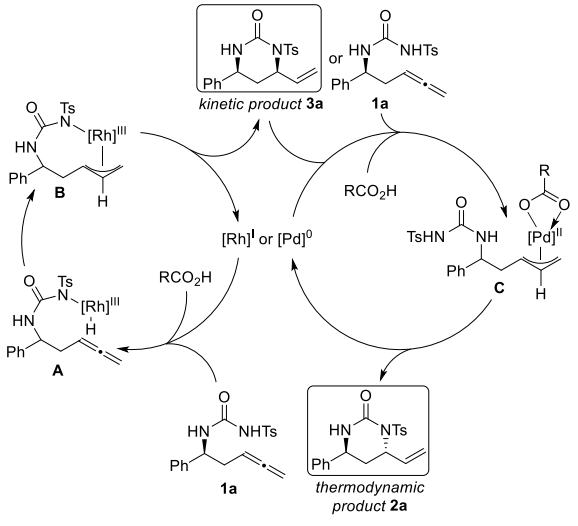
Scheme 4. Control Experiments[c]



[a][Pd(dba)$_2$] (2.0 mol %), DPEphos (5 mol %), Benzoic Acid (50 mol %), DCE (0.2 M), 80 °C, 16 h. [b][{Rh(cod)Cl}$_2$] (4 mol %), DPEphos (10 mol %), Pentafluorobenzoic Acid (20 mol %), DCE (0.2 M), 50 °C, 16 h. [c]DFT-optimized structures with BP86/dev2svp.

In accordance to the proposed mechanisms for the rhodium catalyzed addition of pronucleophiles to allenes and alkynes,[13,18] in particular the catalyzed cyclization of allenyl sulfonylcarbamates,[14] a mechanism for the rhodium catalyzed diastereoselective cyclization of tosylureas has been devised (Scheme 5). For *syn*-selective conditions, electron-deficient carboxylic acids were shown to increase yield and selectivity and may help to generate the active rhodium-hydride intermediate **A** by formation of a suitable precatalyst as was

Scheme 5. Proposed Mechanism for the [Rh]-Catalyzed Intramolecular Addition of Tosylureas to Allenes



https://doi.org/10.1021/acs.orglett.1c03482
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recently described by Heller and Breit.[18b] Hydrometalation of the allene results in the cyclic π-allyl complex **B** which undergoes reductive elimination throughout an inner-sphere reductive elimination to yield the kinetic *syn*-product **3a**. This step seems to be irreversible under the described *syn*-conditions and is favored for strong carboxylic acids, posing weakly coordinating counterions.

Under *anti*-selective conditions, palladium is able to reinsert into the allylic C−N bond and generate the open π-allyl complex **C** in which the catalytic center is saturated with carboxylate. Complex **C** is also intermediate after hydropalladation of allene **1a**. Due to the carboxylate blocking the coordination of the urea functionality, an outer-sphere attack is proposed that favors the formation of the thermodynamically more stable *anti*-product **2a**.

Summarizing we demonstrated the intramolecular addition of tosylureas to allenes to be an efficient method for the diastereoselective preparation of tetrahydropyrimidinones, which are a valuable precursor for the 1,3-diamine motif. Through careful choice of catalyst, both the kinetic *syn*-tetrahydropyrimidinones and thermodynamic *anti*- tetrahydropyrimidinones are accessible in high yields and selectivities. Different follow-up reactions were demonstrated, such as cleavage into the diamine, aldehyde formation, and alkylation. In the future, we are looking to apply our new method in different applications and natural product syntheses.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.1c03482.

Synthetic procedures of new compounds, their analytical data and computational data (PDF)

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was supported by the DFG and Fonds der Chemischen Industrie. The analytical department, in particular Dr. Manfred Keller (Albert-Ludwigs-Universität Freiburg), is acknowledged for help in structural determination. Jasmin Riesterer (Albert-Ludwigs-Universität Freiburg) contributed fundamental research, that we highly appreciate. For his laboratory work and effort, we thank Javad Nisi (Albert-Ludwigs-Universität Freiburg). For contributing DFT calculations we thank Felix Bauer (Albert-Ludwigs-Universität Freiburg).

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https://doi.org/10.1021/acs.orglett.1c03482
Org. Lett. 2021, 23, 9168−9172